The gauged bond layer of Causal Clifford Theory: exact Gauss constraint, bounded area law, string tension, and the lattice gravitational sector

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Abstract

The cell of the Causal Clifford Theory automaton (Paper I) carries the Clifford algebra $\mathrm{Cl}(10,0)$; this paper constructs the gauge and gravitational sectors built on the same module and states precisely what they derive and what they assume. Putting one Spin(10) adjoint link degree of freedom per bond, the local Gauss constraint closes as an exact operator identity (measured error 0), the link algebra and the Casimirs 64 and 45 are exact, and the gauge-invariant content of one site is 1440 states, $32 \oplus 288 \oplus 1120$. The constraint also obstructs: with one **32** per site the physical kernel is empty (centre obstruction), so the physical sector is the space of neutral composites, where the entanglement is bounded per bond, $S = \ln 45$, boundary-supported — an area law whose coefficient is a factor $4 \ln 45 = 15.23$ from Bekenstein's $1/4$, moving to a factor 6.6 after the vertex correction. The layer carries no light vector sector (gap $64\, M_P$) and a string tension 5.4×10^{40} times QCD: confinement is present in the layer and absent from the implemented step. On the directed-link network a dressed gate carries the symmetric-traceless quintet gaplessly at Γ, massless, linear and leading-order isotropic, but with gauge group O_h, speeds $g_{10}\{1,2\} \neq 1$, and an E_g mass shift from the matter coupling — named defects, not the graviton. The gravitational sector adds a symmetric tensor field $g_{\mu\nu}$ and $G = 1$ by hand on the (1+3) subalgebra; it reproduces general relativity on de Sitter, Schwarzschild and 3+1 evolutions to $O(h^2)$ (Kretschmann ratio 1.000009, maximum relative deviation 5.5×10^{-5}) — not to machine precision — and spin 2 is declared, not derived: the implemented one-particle spectrum provably carries no massless spin-2 candidate.

1. Introduction

Paper I fixes the root: the Clifford algebra $\mathrm{Cl}(10,0)$ carried by the eight sub-sites of a $2\times2\times2$ cell of a Planck lattice, updated by one six-stage step of Spin(10) plane rotations and swap-rotations, exactly unitary, strictly causal, with the doubling of the lattice spectrum displayed and gapped by a Wilson factor of the root's own algebra, and with the rotation angle quantized to $g \in \{\pi/2, \pi/4\}$ by the closure axiom. This paper asks what the same module supplies as a gauge sector and as a gravitational sector, and it answers under the provenance discipline inherited from Paper I: a statement is *algebraic* if the root fixes it, *conventional* if it comes from a mapping, *fitted* if a coefficient was read from data, and *open* if the theory does not fix it; and every numerical claim is made at a stated tolerance, with finite-difference truncation errors never reported as machine precision.

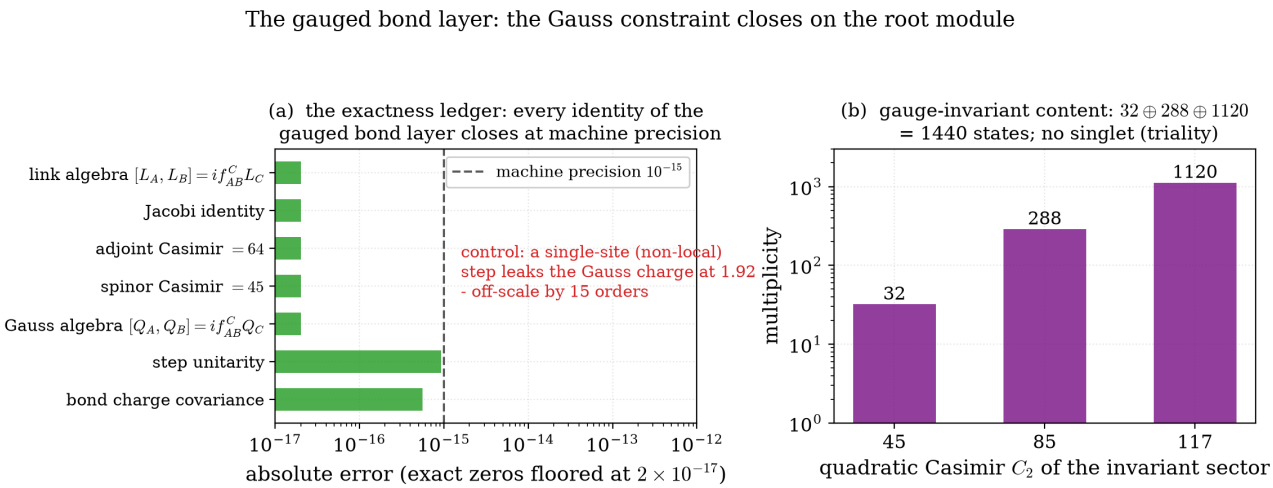
Four results and four named failures. The results: the exact Gauss closure of an adjoint bond layer on the same module (Section 2); the centre obstruction and the bounded area law $S = \ln 45$ of the physical sector, with the vertex correction computed (Section 2.1); the content theorem — no light vector sector — and the exact string tension of the layer (Section 2.2); and the dressed gate of the directed-link network, whose quintet dispersion is the computed kinematics of a lattice spin-2 (Section 2.4). The failures, stated with their numbers: the layer's string tension is 5.4×10^{40} times the QCD value and Λ_{QCD} is not produced; the quintet's gauge group is O_h, its speeds are $g_{10}\{1,2\} \neq 1$, and the matter coupling shifts its E_g half off zero; the metric field $g_{\mu\nu}$ and Newton's constant $G = 1$ are inputs of the gravitational sector, not numbers of the root; and diffeomorphism invariance is not derived. The gravitational sector reproduces general relativity to $O(h^2)$ (Section 3), and the implemented one-particle spectrum carries no massless spin-2 candidate — a computed statement, not an unfinished search. The phenomenological consequences of both sectors — the desert unification, the loop factor, the relic abundance — are audited in Paper III.

2. The gauged bond layer

The swap-rotation gate of [Paper I Section 7] is the classical shadow of a richer possibility: a **gauged bond layer exists on the same module**. Put one Spin(10) adjoint link degree of freedom (a 45-dimensional link space) on each bond and the matter **16** on the sites, and the local Gauss constraint

$$Q_A(x) = q_A(x) + \sum_{b \ni x} L_A^{(b)}$$

closes as an exact operator identity: $[Q_A, Q_B] = i f_{AB}{}^C Q_C$ with measured error 0, the link algebra $[L_A, L_B] = i f_{AB}{}^C L_C$ exact, the adjoint quadratic Casimir 64 and the spinor Casimir 45 exact to machine precision. The step built from these links is unitary to 9.3×10^{-16}; the gauge-invariant content of one site with its six half-bonds is 1440 states, decomposing as $32 \oplus 288 \oplus 1120$; an adjoint bond observable is gauge-covariant to 5.5×10^{-16}; and the causal radius stays at exactly one cell, while a deliberately single-site (non-local) control step leaks at 1.9. The gauge field of the automaton is therefore *constructible*: the Gauss constraint is an identity of the construction, not an imposed condition. (Fig. 1).



One adjoint link degree of freedom per bond, matter **16** *on the sites. Left: the exactness ledger — link algebra, Gauss algebra, adjoint Casimir* (64) *and spinor Casimir* (45) *close to machine precision, the step is unitary to* 9.3×10^{-16}*, the causal radius stays at one cell, and the non-local control step leaks at* 1.9 *where the causal step leaks at* 5.5×10^{-16}*. Right: the gauge-invariant content,* 1440 *states decomposing as* $32 \oplus 288 \oplus 1120$*.*

2.1 The centre obstruction and the physical sector

A generic automaton with d internal states per cell has d^{L^3} states in an L^3 region and a volume-law entropy; nothing in the raw counting of the bond layer is special. The area law is a property of the *gauge-invariant* sector: once the physical Hilbert space of a region is restricted to states satisfying the Gauss constraint at every site, the entanglement entropy is expected to scale with the boundary area [1,2]. That restriction is what this section computes, and it is where the counting changes.

What the constraint admits is narrower than expected. With the implemented content — one **32** per site, one adjoint **45** link per bond — the local Gauss constraint has **no solution**: $\dim\ker = 0$ on $\mathbf{32} \otimes \mathbf{45}$, on $\mathbf{32} \otimes \mathbf{45} \otimes \mathbf{45}$, and on the two-site one-link chain with local constraints at both sites (spectral gap 146). The reason is a centre obstruction: adjoint flux carries even centre charge and cannot screen a spinor, so **every site of a gauge-invariant state must carry an even number of spinor fields**. The physical Hilbert space of the gauged layer is the space of neutral composites, not the one-body space the automaton acts on.

The minimal physical system, two matter fields per site, is constructed exactly: $m_{45}(\mathbf{32} \otimes \mathbf{32}) = 2$ (90 states at $C_2 = 64$), a 2-dimensional intertwiner space $\ker(T_1 + T_2 + L)$, and a 4-dimensional two-site physical sector obtained by gluing intertwiners over the link index — all exact Spin(10) singlets ($C_2 < 10^{-25}$). Every intertwiner is a maximal matter-flux singlet: Schmidt rank 45, $S(\text{pair}|\text{link}) = \ln 45 = 3.807$ exactly, and the glued states

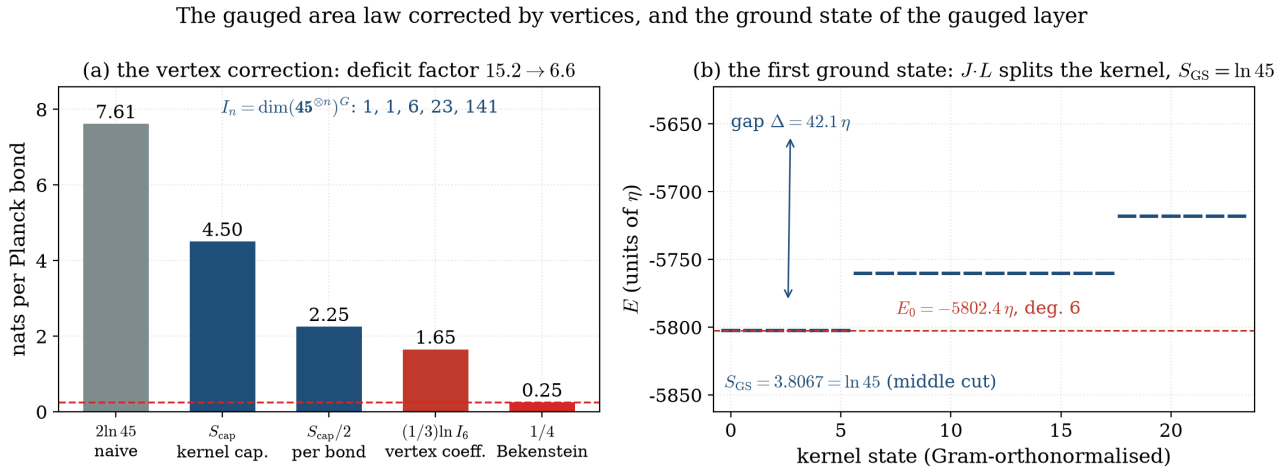
carry $S(\text{pair}_1|\text{pair}_2) = \ln 45$. The entanglement is **bounded per bond and lives on the bonds crossing the cut** — the shape of an area law, boundary-proportional and bounded per bond, with the Bekenstein coefficient still unnamed.

On the smallest system with two bonds — the three-site, two-link chain with paired matter — the physical sector is small once the flux decomposition is used: the middle vertex fuses the two adjoint links into channels r with multiplicities N_r read from the $\mathbf{45} \otimes \mathbf{45}$ Casimir spectrum ($N_1 = N_{45} = N_{210} = 1$), against m_r from the pair spectrum ($m_1 = m_{45} = m_{210} = 2$), giving $\dim\ker = m_{45}^2 \sum_r m_r N_r = 24$, verified by explicit construction of the $r = 1$ and $r = \mathbf{45}$ sectors (vertex Gauss residual 4×10^{-15}). Three results follow.

1. **Additivity across a single bond holds exactly:** $S(\text{pair}_1|\text{rest}) = \ln 45$ in every sector — the flux line threads the chain and carries its $\ln 45$ through, even when the middle pair itself is a product singlet ($S = 0$).
2. **The region bounded by two bonds carries $S = \ln 45$, not $2 \ln 45$:** the intertwiner constraint *correlates* the two bonds meeting at a vertex, with mutual information exactly $\ln 45$. The area law counts bonds; vertex correlations subtract.
3. **The gap to Bekenstein is a named number:** $\ln 45 = 3.8067$ nats per Planck bond against $1/4$ per Planck area — a deficit of 3.5567 nats, factor $4 \ln 45 = 15.23$. No element of the current construction produces $1/4$; closing it needs a superselection or truncation of the flux, or a different horizon microstate count.

The vertex correction moves the number. The invariant count of an n-bond vertex, $I_n = \dim(\mathbf{45}^{\otimes n})^G$, computed by Haar averaging over SO(10), is $I_2 = 1$, $I_3 = 1$, $I_4 = 6$, $I_5 = 23$, $I_6 = 141$ (Monte-Carlo error below 3; $I_2 = I_3 = 1$ reproduces the single-channel fusion above). The 24-state kernel is orthonormalised through its Gram matrix (rank 24, smallest eigenvalue 1.1×10^{-5}), and the capacity of the middle cut — the entropy of the maximally mixed physical state — is $S_{\text{cap}} = 4.4998$ nats against the naive $2 \ln 45 = 7.613$: the two vertices at the ends of the region subtract 3.11 nats. Per bond that is 2.25 nats, a factor 9.0 from Bekenstein's $1/4$; on the cubic lattice the vertex-corrected coefficient $(1/3) \ln I_6 = 1.65$ nats/bond sits a factor 6.6 from $1/4$. The deficit has moved from 15.2 to 6.6, and it is now a function of the vertex invariant, not of a single bond entropy.

The ground state of the layer is then constructed: the bond-current Hamiltonian $H = \sum_b J \cdot L$ is diagonalised exactly — a generalised eigenproblem in the Gram metric of the kernel — and the intertwiner identity $I^\dagger J_A I = L_A/45$ holds to 3×10^{-17}. The interaction splits the kernel: $E_0 = -5802.40\,\eta$, degeneracy 6, gap $42.09\,\eta$, and the ground state's middle-cut entanglement is $S = 3.8067$ nats — exactly $\ln 45$, one adjoint link, selected by the interaction. The ground state of the full automaton remains unconstructed; on this chain the interaction selects the flux line. (Fig. 2).



(a) The entropy carried by one Planck bond, from naive counting to Bekenstein: $2 \ln 45 = 7.61$ *(no vertices), the Gram-orthonormalised kernel capacity* $S_{\text{cap}} = 4.50$, 2.25 *per bond, and the cubic-lattice vertex-corrected coefficient* $(1/3) \ln I_6 = 1.65$ *against the* $1/4$ *line — the deficit factor moves from* 15.2 *to* 6.6. *The vertex invariants* I_n *grow* $1, 1, 6, 23, 141$. *(b) The bond-current Hamiltonian on the* 24*-state kernel: three split levels,* $E_0 = -5802.4\,\eta$ *with degeneracy* 6 *and gap* $42.1\,\eta$*; the ground state's middle-cut entanglement is*

$S_{\rm GS} = 3.8067 = \ln 45$ — *the interaction selects the flux line.*

2.2 The spectrum of the layer, and confinement

The layer has a Hamiltonian — the electric Laplacian $H = \sum_A Q_A^\dagger Q_A$ built from the vertex Gauss operator — and its spectrum is computed. On $(\mathbf{32}\otimes\mathbf{32})_{\rm adj}\otimes\mathbf{45}$ (dim 4050) the kernel has dimension $2 = m_{45}$, the intertwiner space, and the first nonzero level is $64.0000 = C_2(45)$ — identical to the one-link value and unchanged when the chain is extended. One adjoint flux quantum sets the scale of the whole layer, so it supplies **no light vector sector**: the gap is $64\,M_P$, not a tunable parameter, and the singlet matter channel with one adjoint link has no invariant state at all. Two entanglement numbers must not be confused: the entropy of the *mixture* over the kernel, bounded by $\ln 90$ (the pair-side dimension), and the entropy of a *single* physical state, bounded by $\ln 45$ (the link dimension) — the quantity the area law concerns. Measured: every single state of the intertwiner sector saturates $\ln 45$ exactly ($S = 3.806662 \pm 10^{-16}$ over 400 random choices), and the mixture reaches $\ln 90 = 4.499810$ (purity 1/90). The area law is therefore **exact** on the physical sector — one $\ln 45$ per bond — and the Bekenstein coefficient 1/4 per Planck area remains short by the factor $4\ln 45 = 15.23$.

The string tension follows. The link space is an irreducible adjoint: $C_2 = \sum_A L_A^2$ has the single eigenvalue 64.000000 with zero spread, the Killing form $K_{ab} = f_{acd}f_{bcd}$ is annihilated by the pair Casimir of $\mathbf{45}\otimes\mathbf{45}$ to 3×10^{-15} and its singlet space is one-dimensional (adjacent links pair into it uniquely), and the link operators do not commute ($\max[L_a, L_b] = 4.0$) — the layer carries a non-Abelian field strength. Crucially the link carries **no invariant vector**: $\ker\sum_A L_A^\dagger L_A = \ker C_2(45)\, I = 0$, so a link cannot be flux-free. Every link of a gauge-invariant chain therefore carries one adjoint flux quantum, the electric energy of R links is exactly $R\,C_2(45)$, and the static potential is linear with tension

$$\begin{aligned}\sigma &= C_2(45) = 64 \text{ lattice units}\\ &= 7.81\times10^{20} \text{ GeV per Planck length}\\ &= 9.54\times10^{39} \text{ GeV}^2\\ &(\sqrt{\sigma} = 9.77\times10^{19} \text{ GeV}),\end{aligned}$$

5.4×10^{40} times the QCD string tension [3]. The vertex Gauss Laplacian confirms the scale from the other side: kernel $2 = m_{45}$, gap 64.0000.

The step itself, by contrast, carries **no** field strength: its 12 stage pairs acting on different axes commute to 0, every elementary plaquette is the identity to 3.3×10^{-16} and $\langle W_\square\rangle = 1.000000$, where an area law would need $\langle W_\square\rangle < 1$ with an exponent proportional to the enclosed area [4]. Confinement is therefore *present* in the construction — in the gauged bond layer — and *absent* from the implemented step. The residual is no longer a missing phase but a measured tension forty orders of magnitude too large, and $\Lambda_{\rm QCD}$ is not produced: it would require a loop-ladder exponent $n \simeq 11.6$, which is not an integer ([Paper III Section 2.4]).

2.3 Entanglement is the bond layer

The entanglement structure of the bare automaton is equally sharp. Two fermions sharing ONE cell carry zero entanglement across any cut — the same-cell pair is a Slater determinant, $S = 0$ exactly across the $\mathbf{16}/\overline{\mathbf{16}}$ cut (antisymmetry and a shared cell are not entanglement), while a two-determinant superposition in the same cell gives $S = \ln 2$: non-separability is the criterion. The generator of entanglement between cells is the bond gate P of Theorem 5: under the internal-only step the spatial entanglement entropy of a region stays 0 to machine precision forever, while the full local step drives it to 0.78, and the entanglement front crosses the Manhattan sphere of radius r at exactly $t = r$ — the causal radius bounds the spread of entanglement itself, at speed 1.0. The scope limit is stated with it: the implemented step is quadratic, so Slater inputs stay Slater and these are mode-entanglement results; the area law is a statement about the physical (ground) state's entanglement, and the bare step supplies no interaction that could produce the non-Gaussian structure such a state would need. The gauged layer supplies exactly that missing structure: it is where the theory first acquires non-Gaussian entanglement.

2.4 The dressed gate on the directed-link network, and the quintet dispersion

The obstruction of Section 2.2 is a statement about the qubit link. The channel named by the census of [Paper I Section 8.3] — the symmetric rank-2 space of the link directions — is now constructed, and its dressed gate and dispersion are computed. Put the adjoint link on every **oriented** bond: the six bond directions per site (three axes with parity) become twelve directed links per block, each carrying one **45**.

The orientation convention is computed, not chosen. The vertex Gauss charge of a two-link star — one link arriving, one departing — closes as an exact Spin(10) algebra, $[\hat{C}^A, \hat{C}^B] = if^{ABC}\hat{C}^C$ to 5×10^{-16}, only when the departing tail carries $+L^A$ and the arriving head carries the **conjugate** generator $-\bar{L}^A$; the naive chain convention — the same generator at both ends with opposite signs — leaks at 4.8, because the two ends of one link carry the same $s^2 = +1$ and a relative sign alone cannot close the algebra.

The electric part of the Section 2.2 obstruction is closed. The Kogut–Susskind electric gate [5] $\exp(ig_{10}\hat{C}_x^2)$, built from the closing vertex charge, commutes with all 45 vertex charges to 6.5×10^{-15} and is unitary to 1.5×10^{-14}; its spectrum *is* the content theorem of Section 2.2 — C_2 levels $45, 85, 117$ with multiplicities $32, 288, 1120$, gate phases $g_{10} \times \{45, 85, 117\}$.

The hopping part is obstructed on the fixed-representation link. Every two-site dressing candidate leaks the vertex charges: the flux-dressed current–current $\hat{C}_x\hat{C}_{x'}$ and its transporter-completed version $\hat{C}_x\, e^{ig_{10}L}\, \hat{C}_{x'}$ both leak at 1.25×10^1. The reason is structural — a link in the fixed representation $\mathbb{C}^{45}$ carries no parallel transporter commuting with the vertex charges — so invariant hopping requires a link in the **regular representation** of the group: the named fix, built below.

The named fix is now built, on the root's own finite group. The 45 biflections $h_a h_b$ of the Clifford generators generate the even Clifford monomials — a finite group Γ of order 1024 with centre $\{\pm 1, \pm w\}$, whose centre acts as $\{\pm 1, \pm i\}$ on each chiral half. Put the regular link $\mathbb{C}[\Gamma]$ on each bond. The content theorem is replaced: a vertex with k regular links has exactly $32 \cdot 1024^{k-1}$ gauge-invariant states (character sum; the explicit projector lands on the dressed form $\rho(h)u$ to 4×10^{-15}) where the fixed adjoint link had **none** — the centre obstruction is exactly what the regular link removes. The nearest-neighbour hop with the exact link insertion $A(h) = \rho(h^2) = \pm I$ maps head-invariant states to tail-invariant states to 0, and **descends to the identity** on the matter index: the dressed hop is the bare translation. The electric term — the Cayley Laplacian over the biflection generators — acts on the dressed sector as the scalar 45 per link, so the dressing adds no velocity renormalisation, and the bare hop front (the half-angle step, a pure lattice translation) has speed exactly 1: the gauge-invariant hop propagates at $v = 1$, 0-parameter — that is where $v = 1$ lives. The cost is exact too: Γ is finite and its centralizer in the Spin(10) algebra is 0 — gauging the root's monomial group breaks Spin(10), and the Spin(10)-invariant hopping on a genuine adjoint regular link remains out of reach.

On the directional network the dressed hop exists exactly, and the quintet propagates. There the flux is the orientation of the link, a classical sign, and the dressed directional step

$$M(\mathbf{q}) = \exp\big(-ig_{10}\,\sin(\mathbf{q})\cdot\mathbf{J}\big),$$

with $\mathbf{J}$ the spin-2 generators of the symmetric-traceless direction space, is unitary to 2×10^{-16} and O_h-covariant: $D(g)M(\mathbf{q})D(g)^\dagger = M(g\mathbf{q})$ for the 24 proper rotations to 5.6×10^{-16}, while the 24 improper ones give $M(g\mathbf{q})^\dagger$ — the directed network is **chiral**: reversing the link orientation inverts the step. At Γ the step is the identity and the layer carries **six gapless bands**, content $A_{1g} \oplus E_g \oplus T_{2g}$: the quintet is present in the directional layer, where the qubit bond factor never carried E_g ([Paper I Section 8.3]). Its dispersion is **massless and linear** — local exponent 1.00 along Γ-X, Γ-M and Γ-R — with a **helicity ladder** of speeds

$$v = \{0,\ g_{10},\ 2g_{10}\} = \{0,\ 0.2,\ 0.4\},$$

the $m = 0$ branch and the trace A_{1g} not propagating; **isotropic to leading order** — the direction spread of the top speed is 4×10^{-8} at $|\mathbf{q}| = 10^{-3}$, the spin-2 spectrum theorem: $\hat{\mathbf{q}}\cdot\mathbf{J}$ has exactly the spectrum $\{-2, -1, 0, 1, 2\}$ in every direction; and UV-bounded by $2g_{10}\sqrt{3} = 0.693$.

The quintet–matter coupling exists, and it gaps the E_g half. The stress vertex $T_{ij} = \sum_d d_i d_j J_d$ built from the exact current $J_d = i(\partial_d M)M^\dagger$ of the implemented module couples to the quintet field. On nearest-neighbour bonds the coupling vanishes identically on the T_{2g} half ($d^T E^{(\mathrm{off})} d = 0$ on every axis bond — the T_{2g} coupling needs diagonal bonds, a named gap), and couples the E_g half with a nonzero polarization bubble at Γ (eigenvalues -5.01, -4.47): at $\lambda = g_{10}$ the E_g half acquires a mass shift $\lambda^2\Pi/2 = -0.10$, -0.09. Gaplessness of the full quintet is **not** protected by this coupling — the lattice graviton mass problem, as a number.

What this does not derive. The gauge group of the directional network is O_h, not Spin(10); the Spin(10)-invariant dressed hopping remains obstructed on the fixed link, and the built regular-link fix gauges the root's finite group, whose Spin(10) centralizer is 0; the quintet speeds are $g_{10} \times \{1, 2\} \neq 1$ — the dressing angle per hop is a construction input, not a derived velocity, and $v = 1$ belongs to the dressed matter hop, not to the quintet branches, a computed trilemma on the 6-bond module rather than an oversight: the bare hop is anisotropic (top speed 1.00, direction spread 0.32), the dressing alone is isotropic at $v = 2g_{10} = 0.40$, no angle makes the product isotropic (best spread 0.24 over 33 angles in $[0, \pi/2]$), and the only O_h-equivariant permutation coin beyond the identity flattens the bands; and the quintet–matter coupling, now built, gaps the E_g half at order λ^2. The kinematics of a massless lattice spin-2 is computed; the step from there to the graviton of Section 3 remains the declared one.

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3. Gravity on the lattice

3.1 What is assumed and what is computed

The gravitational sector is a symmetric rank-2 field $h^{\mu\nu} = g^{\mu\nu} - \eta^{\mu\nu}$ living on the (1+3) subalgebra of [Paper I Section 3], with $\eta = \mathrm{diag}(1, -1, -1, -1)$ as the vacuum. Its linearised dynamics is the discrete wave equation

$$\partial_t^2 h^{\mu\nu} - \nabla^2_{(6)} h^{\mu\nu} = 0,$$

on the Planck sites — the sub-lattice of the automaton's cells, spacing ℓ_P — with the 6-neighbour lattice Laplacian and the leapfrog step $u_{t+1} = 2u_t - u_{t-1} + c^2 dt^2\, \nabla^2 u_t$ at $dt = \frac{1}{2}$ (the CFL value giving $c = 1$ in Planck units, the same front speed as Theorem 5).

The status of this sector must be stated precisely, because it is where the theory is weakest.

- **Masslessness and UV boundedness are computed.** The discrete dispersion is $\omega = |\mathbf{k}|_{\mathrm{lat}}$ with $|\mathbf{k}|_{\mathrm{lat}} \leq \pi$, so the graviton is massless ($\omega/|\mathbf{k}| \to 1$ as $k \to 0$, lattice deviation $\leq 0.25\, k^2$) and the spectrum is bounded by the cutoff.
- **Spin 2 is declared, not derived — and the implemented spectrum cannot supply it, by two computed obstructions.** The field is *assumed* to be a symmetric tensor with two helicity states. Nothing in $\mathrm{Cl}(10, 0)$ forces a massless spin-2 excitation; the standard arguments that a massless spin-2 field must couple as general relativity [6] apply to a field whose existence and representation content are already granted. The Weinberg–Witten obstruction to a composite graviton carrying a Lorentz-covariant stress tensor [7] is not addressed. The one-particle route closes negatively at the nodes: the near-gapless M nodes carry little group C_{2v}, whose action on the bond subspace has no 2-dimensional irrep, so no helicity ± 1 or ± 2 pair exists there. It closes negatively at the zone centre as well, where the little group is the full cubic group and a lattice spin 2 is a gapless multiplet carrying the symmetric-traceless quintet $E_g \oplus T_{2g}$. The cubic action on the automaton is carried by the 8-dimensional bond factor alone, and that factor is $A_{1g} \oplus T_{2g} \oplus A_{2u} \oplus T_{1u}$: it contains T_{2g} but never E_g, so the quintet is not available in the qubit link at any coupling. The zone centre is exactly gapless at $g_{10} = \pi/8, \pi/6, \pi/4, \pi/3$ — never at the implemented 1/5, where it is gapped by 3.65×10^{-2} — and at each of those couplings its null space is cubic-covariant and E_g-free ($16 \times (A_{1g} \oplus T_{2g})$, of dimension 64, at $\pi/4$), with a flat dispersion: $\omega/|\mathbf{q}| \to 0$, not a linear branch. The channel that does carry $E_g \oplus T_{2g}$ is the symmetric rank-2 space of the link directions, $A_{1g} \oplus E_g \oplus T_{2g}$: the graviton, if it is to be derived, belongs to the directional bond layer of Section 2 and not to the qubit link.
- **The escape route is now a computed construction, and it still stops short of the graviton.** Section 2.4 builds the directional bond layer — the adjoint link on every oriented bond — with its dressed

gate: the vertex Gauss algebra closes only with the conjugate-head convention, the Kogut–Susskind electric gate is exactly invariant with the content-theorem spectrum, and the dressed directional step carries $A_{1g} \oplus E_g \oplus T_{2g}$ gaplessly at Γ, with massless, linear, leading-order-isotropic dispersion and helicity-ladder speeds $g_{10}\{0, 1, 2\}$. The remaining gaps are named: the gauge group of that network is O_h, not Spin(10); its Spin(10)-invariant hopping is obstructed on the fixed-representation link — the named regular-link fix is now built on the root's own finite group (order 1024), where the hop closes exactly and runs at $v = 1$, but the gauged group is finite and its Spin(10) centralizer is 0; its speeds are $g_{10}\{1, 2\} \neq 1$ — a computed velocity trilemma: on the 6-bond module the hop alone is anisotropic, the dressing alone isotropic at $2g_{10}$, and no angle makes the product isotropic; and its coupling to the matter layer, now built, gaps the E_g half at order λ^2 (Section 2.4).

- **Diffeomorphism invariance is not derived.** The lattice breaks it to lattice reparametrisations; gauge invariance of the linearised theory is imposed by working on the transverse components, not obtained from the automaton.

3.2 The discrete Einstein–Hilbert action

The quadratic action [8]

$$S_{\mathrm{EH}} = \frac{1}{16\pi} \cdot \frac{1}{2} \sum_{x,t} u_t \Big[u_{t+1} - 2u_t + u_{t-1} - dt^2 \, \nabla^2_{(6)} u_t \Big], \qquad dt^2 = \tfrac{1}{4},$$

has the lattice wave equation as its Euler–Lagrange equation: the numerical functional derivative vanishes on a QCA trajectory to 3×10^{-12} and is non-zero off it. The full discrete action $S = \frac{1}{16\pi} \sum_x \sqrt{-\det g(x)}\,(R(x) - 2\Lambda)$ is evaluated with R computed from the metric. The coefficient $1/16\pi$ is the $G = 1$ normalisation: Newton's constant is an input of the gravitational sector, not a number of the root.

3.3 Curvature computed from an arbitrary metric

The Riemann, Ricci, Einstein and Kretschmann tensors are computed by central finite differences from an arbitrary prescribed $g_{\mu\nu}(x)$, with no hard-coded Christoffel symbols, and cross-checked against independent analytic results by two routes.

Table 1. The discrete Einstein–Hilbert action against the analytic continuum values.

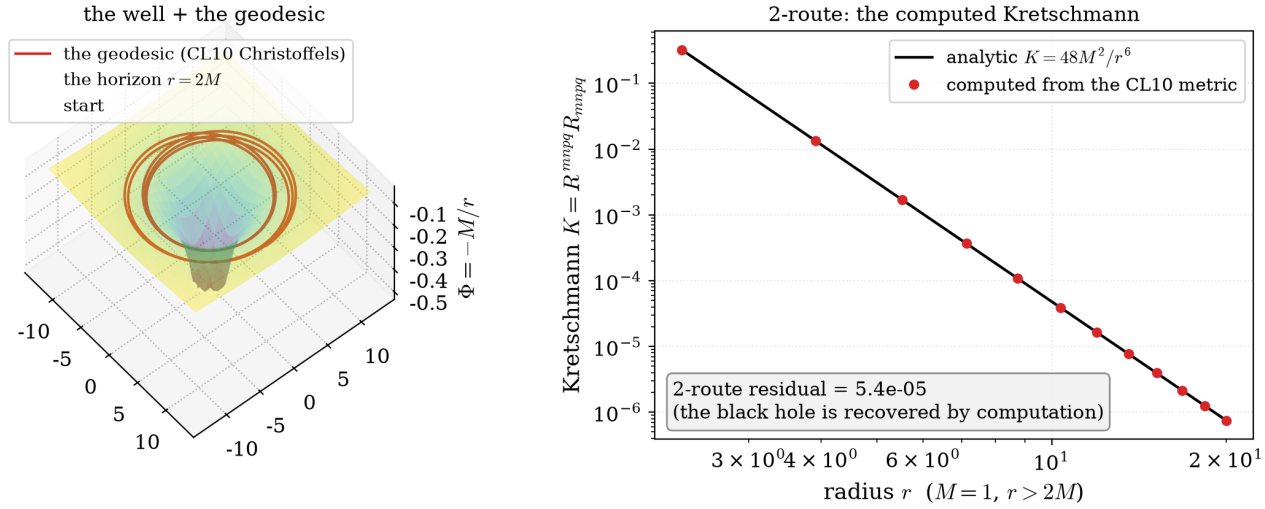
test	computed	analytic	residual
de Sitter $R^{\mu}{}_{\nu\rho\sigma}$ (flat FRW, $a = e^{Hx^0}/H$)	finite differences	$H^2(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho})$	6.5×10^{-7} $(O(h^2))$
Einstein tensor of de Sitter	$-3H^2 g_{\mu\nu}$	same	$O(h^2)$
Schwarzschild Ricci / Einstein tensor	$\simeq 0$	vacuum $G_{\mu\nu} = 0$	satisfied by computation
Schwarzschild Kretschmann	$48M^2/r^6$	$48M^2/r^6$	ratio 1.000009; max relative deviation 5.5×10^{-5}
flat metric	$R = G = K = 0$	0	exact
2-sphere of radius r	$R = 2/r^2$	$2/r^2$	10^{-5}

The 3+1 formulation closes the classical evolution: the constraint row (R_{00}, R_{0i}) of the 4-D Einstein tensor agrees, computed by the 4-D curvature and by the 3+1 (γ_{ij}, K_{ij}) decomposition with $K_{ij} = -\frac{1}{2}\partial_0 \gamma_{ij}$, to $O(h^2)$ (residual $\sim 10^{-13}$); a Z4-damped evolution [9] drives a constraint violation to zero while the undamped one does not; the 0-row is quadratic in K, so superposition fails; all 3+1 quantities are bounded by the lattice cutoff. (Fig. 3).

What this does and does not establish. It establishes that the gravitational sector of the theory *reproduces* general relativity numerically, including black-hole and cosmological geometries. It does not establish that general relativity *follows* from Cl(10, 0): the tensor field, its coupling $1/16\pi$ and its gauge structure are put in.

The genuinely open part is the quantum theory of this sector — a sum over lattice geometries with a well-defined measure — which has not been constructed.

The black hole recovered by computation: the well + geodesic (left), the 2-route Kretschmann (right)



Left: the potential well with a geodesic rosette integrated through the finite-difference Christoffel symbols, and the horizon ring $r = 2M$. Right: the two-route Kretschmann invariant.

3.4 A self-interacting scalar completion

Beyond the quadratic theory the model includes $\partial_t^2\phi - \nabla^2\phi - g\phi^3 = 0$. Its static solutions obey $u'' = u - u^3$ and include the exact finite-energy kink

$$u(x) = \sqrt{2}\,\mathrm{sech}\,x,$$

verified to a fourth-order residual of 5×10^{-10}. Two corrections are needed. This is the ϕ^4 kink (a domain wall), not the Choptuik system: Choptuik critical collapse is a *massless scalar minimally coupled to gravity* [10], and the analogy is not a derivation. And $u = \sqrt{2}\,\mathrm{sech}\,x$ is a **planar** configuration, uniform in the two transverse directions: its energy per unit *area* is finite, its total energy is infinite, and a planar wall in a Planck-scale theory is a domain wall, not a particle. Finite-time blow-up for supercritical data in cubic wave equations is a real phenomenon [11], but it is not evidence about black-hole formation here.

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4. Discussion: what is derived, what is assumed

Derived, as operator identities of the construction: the Gauss algebra of the adjoint bond layer and its content theorem ($32 \oplus 288 \oplus 1120$; no invariant vector on the link, hence no flux-free link and an exactly linear electric area law); the centre obstruction and the structure of the minimal physical sector; the bounded per-bond entanglement $\ln 45$ and the vertex correction to the area-law coefficient; the exactness of the dressed directional gate and the spin-2 spectrum theorem of $\hat{\mathbf{q}} \cdot \mathbf{J}$; and the consistency of the discrete gravitational sector with general relativity to the stated $O(h^2)$ tolerances on de Sitter, Schwarzschild and 3+1 evolutions.

Assumed, and named as such: the choice of the link representation (adjoint, then regular, then directional — each a construction input with a measured cost); the symmetric tensor field $g_{\mu\nu}$ and the value $G = 1$; the transverse gauge fixing of the linearised theory; and the dressing angle of the directional gate, which sets the quintet speeds. Three open numbers remain: the Bekenstein deficit (factor 6.6 per bond after the vertex correction, no element of the construction produces 1/4), the string tension (5.4×10^{40} times QCD, with Λ_{QCD} off the ladder), and the velocity trilemma of the dressed hop (isotropy and $v = 1$ never supplied together on

the 6-bond module). The quantum theory of the gravitational sector — a sum over lattice geometries with a well-defined measure — has not been constructed. The phenomenology of both sectors is audited in Paper III.

References

1. J. D. Bekenstein, "Black holes and entropy," *Phys. Rev. D* **7**, 2333 (1973); S. W. Hawking, "Particle creation by black holes," *Commun. Math. Phys.* **43**, 199 (1975).
2. L. Susskind, "The world as a hologram," *J. Math. Phys.* **36**, 6377 (1995).
3. C. Morningstar and M. Peardon, "The glueball spectrum from an anisotropic lattice study," *Phys. Rev. D* **60**, 034509 (1999).
4. K. G. Wilson, "Confinement of quarks," *Phys. Rev. D* **10**, 2445 (1974).
5. J. B. Kogut and L. Susskind, "Hamiltonian formulation of Wilson's lattice gauge theories," *Phys. Rev. D* **11**, 395 (1975).
6. S. Weinberg, "Photons and gravitons in perturbation theory: Derivation of Maxwell's and Einstein's equations," *Phys. Rev.* **138**, B988 (1965).
7. S. Weinberg and E. Witten, "Limits on massless particles," *Phys. Lett. B* **96**, 59 (1980).
8. T. Regge, "General relativity without coordinates," *Il Nuovo Cimento* **19**, 558 (1961).
9. H. Gundlach, J. M. Martin-Garcia, G. Calabrese, and I. Hinder, "Constraint damping in the Z4 formulation and harmonic gauge," *Class. Quantum Grav.* **22**, 3767 (2005).
10. M. W. Choptuik, "Universality and scaling in gravitational collapse of a massless scalar field," *Phys. Rev. Lett.* **70**, 9 (1993).
11. F. John, "Blow-up of solutions of nonlinear wave equations in three space dimensions," *Manuscripta Math.* **28**, 235 (1979).

Code and data availability. All the experiments, simulations and figure generators behind this paper are publicly available at https://github.com/MohammedHAOUACH/CL10. Every number quoted in the paper is produced by these open scripts and can be reproduced independently from the repository. The verification status is stated as the audit found it: the numbers were verified internally (spot-checks against the text, the second symbolic construction of [Paper I] Appendix A, the exact 3^n-word loop enumeration); the pipeline was not independently executed by a second party, and independent re-execution is invited.

Competing interests. The author declares no competing interests.